Double Queue for Constrained Online Convex Optimization: Bridging the Best-of-Two-Worlds Constraint Violations

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Abstract*—We study Constrained Online Convex Optimization (COCO) in the two worlds of time-varying and time-invariant constraints. Our goal is to simultaneously minimize the regret to the best fixed solution in hindsight and the hard violation prohibiting any compensated constraint violations. We name the proposed COCO algorithm DOUBLE-Q, since it leverages both an original virtual Queue to construct a surrogate loss and a surrogate virtual Queue to construct a surrogate drift-plus-penalty (SDPP). This new double-queue design enables tighter dual violation control compared to existing single-queue approaches. By minimizing an upper bound on the SDPP at each time, DOUBLE-Q achieves $\mathcal{O}(\sqrt{T})$ regret and $\mathcal{O}(\min\{\sqrt{T}\log T, T^v\})$ hard violation without requiring Slater's condition, where v continuously quantifies constraint variation from 0 for fixed constraints to 1 for arbitrary constraints. For the first time, this hard violation bound simultaneously guarantees the best-known $\mathcal{O}(\sqrt{T}\log T)$ violation for arbitrary constraints at worst, and recovers the best-known $\mathcal{O}(1)$ violation for fixed constraints at best, thus bridging the best-known constraint violations in the two COCO worlds. For strongly convex loss functions, DOUBLE-Q achieves both improved $\mathcal{O}(\log T)$ regret and $\mathcal{O}(\min\{\sqrt{T\log T}, T^v\})$ hard violation. Simulation results demonstrate that DOUBLE-Q outperforms state-of-the-art COCO algorithms across various online applications.

I. INTRODUCTION

Online Convex Optimization (OCO) is a vital framework for sequential decision-making under uncertainty [1]-[3]. The standard OCO models a sequential game between a learner and an adversary over time. At each time, the learner first selects a decision from a bounded convex feasible set. The adversary then reveals a convex loss function, and the learner incurs the corresponding loss. Faced with unknown loss functions, the learner cannot achieve the optimal performance. Instead, the learner aims to minimize the *regret*, which quantifies the cumulative performance gap between its online decisions and the best fixed decision in hindsight. The seminal OCO work [4] achieved $\mathcal{O}(\sqrt{T})$ regret for a sequence of T arbitrary convex loss functions (with bounded gradients). The regret was improved to $\mathcal{O}(\log T)$ for strongly convex loss functions [5].

The standard OCO requires constraints to be strictly satisfied at each time, which often involves computationally expensive projection operations onto the feasible set [4]-[9]. Furthermore, many practical networking applications [10]-[13] only require constraints to be satisfied cumulatively over time. These motivate constrained OCO (COCO) [14]-[27], where some constraints in the feasible set are relaxed to long-term constraints. In COCO, the learner aims to simultaneously minimize the regret and the cumulative *constraint violation* over time.

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Early COCO works focused on *known time-invariant* constraints [14]-[19], proposing two constraint violation metrics: *soft* violation allowing compensated violations [14] and *hard* violation prohibiting any compensated violation over time [17]. The hard violation has attracted more attention since a bound on hard violation [17]-[19] readily implies a bound on soft violation [14]-[16]. For *fixed* constraints, state-of-the-art COCO algorithms achieved both $\mathcal{O}(1)$ soft [16] (requiring Slater's condition) and hard violation [19].

Recent COCO works studied *unknown time-varying* constraints [19]-[23], where both the loss and constraint functions are revealed only after the learner selects its decision. This adversarial setting inevitably incurs constraint violations. For *arbitrary* convex constraints (with bounded gradients and/or values), state-of-the-art COCO algorithms achieved $\mathcal{O}(\sqrt{T})$ soft violation [21] (requiring Slater's condition) and $\mathcal{O}(\sqrt{T}\log T)$ hard violation [22]. However, compared with the best-known $\mathcal{O}(1)$ violations for fixed constraints in the time-invariant COCO world, the best-known violations for arbitrary constraints increase significantly to $\Omega(\sqrt{T})$ in the time-varying COCO world.

To bridge the above two COCO worlds, some works quantify the cumulative constraint variation over time, enabling a *unified* constraint violation analysis for both time-invariant and time-varying COCO [24]-[27]. Along this direction, the state-of-the-art algorithm [27] achieved $\mathcal{O}(T^v)$ hard violation for both convex and strongly convex loss functions, where v represents the constraint variation (from 0 for fixed constraints to 1 for arbitrary constraints). This hard violation bound recovers the $\mathcal{O}(1)$ violation achievable for fixed constraints in the time-invariant COCO world *at best*. However, compared with the $\mathcal{O}(\sqrt{T}\log T)$ hard violation achievable for arbitrary constraints [22], this bound degrades to $\mathcal{O}(T)$ in the time-varying world *at worst*.

The significant performance gap between time-invariant and time-varying COCO worlds leads to a fundamental question: *Can a COCO algorithm provide a hard violation bound that guarantees i) $\mathcal{O}(\sqrt{T}\log T)$ violation for arbitrary constraints at worst, and at the same time, ii) recovers $\mathcal{O}(1)$ violation for fixed constraints at best, thus bridging the best-known hard violations for time-varying and time-invariant COCO worlds?* In this work, a new *double-queue* design generates two complementary violation bounds: one robust for arbitrary constraints and one tight for fixed constraints. Their minimum achieves both i) and ii) simultaneously, contrasting with existing single-queue approaches limited to achieving either one. This leads to a hard violation bound that is *never worse* and can be *strictly better* than state-of-the-art COCO algorithms for *any* constraint violation (see detailed comparisons in Table I).

We summarize our main contributions below:

• We propose an efficient algorithm for COCO, handling both time-varying and time-invariant constraints. We name the algorithm DOUBLE-Q, since it leverages both an original virtual Queue to construct a surrogate loss and a surrogate virtual Queue to construct a surrogate drift-plus-penalty (SDPP). Unlike existing single-queue COCO approaches, this new double-queue design provides tighter dual control over the constraint violations, even without requiring Slater's condition.

• We analyze the performance of DOUBLE-Q, which minimizes an upper bound of the SDPP at each time. Our analysis shows that DOUBLE-Q achieves $\mathcal{O}(\sqrt{T})$ regret and $\mathcal{O}(\min\{\sqrt{T}\log T, T^v\})$ hard violation, where v continuously quantifies the cumulative constraint variation (see its definition in (2)). Notably, DOUBLE-Q is the first to simultaneously guarantee $\mathcal{O}(\sqrt{T}\log T)$ hard violation for arbitrary constraints at worst ($v = 1$) and recover $\mathcal{O}(1)$ hard violation for fixed constraints at best ($v = 0$), even without knowing v.

• We further analyze the performance of DOUBLE-Q under strongly convex loss functions. Our analysis shows that DOUBLE-Q achieves both improved $\mathcal{O}(\log T)$ regret and $\mathcal{O}(\min\{\sqrt{T\log T}, T^v\})$ hard violation. Notably, DOUBLE-Q remains the first to simultaneously guarantee $\mathcal{O}(\sqrt{T\log T})$ hard violation at worst and recover $\mathcal{O}(1)$ hard violation at best for COCO with strongly convex loss functions.

• We conduct numerical experiments to evaluate the performance of DOUBLE-Q on various applications including online quadratic programming, online job scheduling, and online fraud detection, with both convex and non-convex losses under real-world datasets. Simulation results demonstrate that DOUBLE-Q consistently provides superior performance over state-of-the-art baselines.

II. RELATED WORK

The seminal OCO work [4] achieved $\mathcal{O}(\sqrt{T})$ static regret to the best fixed decision. Subsequent work [5] improved the static regret to $\mathcal{O}(\log T)$ for strongly convex loss functions. A notion of dynamic regret to a sequence of time-varying decisions was also introduced in [4]. Subsequent OCO works developed various algorithms to bound the dynamic regret [6]-[9]. In this work, we focus on bridging the best achievable hard violations in time-invariant [14]-[19] and time-varying COCO [19]-[23], under the static regret (regret for short).

A. Time-Invariant COCO

Early COCO works focused on known time-invariant constraints and adopted the soft violation metric [14]-[16], which allows compensated constraint violations over time. The saddle-point-type algorithm in [14] — constructing Lagrangian functions followed by primal-dual gradient descent — achieved $\mathcal{O}(\sqrt{T})$ regret and $\mathcal{O}(T^{\frac{3}{4}})$ soft violation. Subsequent work [15] achieved $\mathcal{O}(\max\{T^c, T^{1-c}\})$ regret and $\mathcal{O}(T^{1-\frac{c}{2}})$ soft violation, where $c \in [0, 1]$ is a trade-off parameter. The virtual-queue-based algorithm in [16] — constructing virtual queues followed by drift-plus-penalty upper bound minimization — achieved the best-known $\mathcal{O}(1)$ soft violation under Slater's condition, which excludes important equality constraints in many practical applications.

TABLE I: COCO performance bounds without Slater's condition for any constraint variation $v \in [0, 1]$ (0: fixed constraints; 1: arbitrary constraints).

Algorithm	Loss Function	Regret	Hard Violation
[19]	convex	$\mathcal{O}(\sqrt{T})$	$\mathcal{O}(T^{3/4})^*$
[22]	convex	$\mathcal{O}(\sqrt{T})$	$\mathcal{O}(\sqrt{T}\log T)^\dagger$
[27]	convex	$\mathcal{O}(\sqrt{T})$	$\mathcal{O}(T^v)^\ddagger$
DOUBLE-Q	convex	$\mathcal{O}(\sqrt{T})$	$\mathcal{O}(\min\{\sqrt{T}\log T, T^v\})$
[19]	strongly convex	$\mathcal{O}(\log T)$	$\mathcal{O}(\sqrt{T\log T})$
[22]	strongly convex	$\mathcal{O}(\log T)$	$\mathcal{O}(\sqrt{T\log T})$
[27]	strongly convex	$\mathcal{O}(\log T)$	$\mathcal{O}(T^v)$
DOUBLE-Q	strongly convex	$\mathcal{O}(\log T)$	$\mathcal{O}(\min\{\sqrt{T\log T}, T^v\})$

* For fixed constraints ($v = 0$), the hard violation bound in [19] is $\mathcal{O}(1)$. DOUBLE-Q is better in almost all variations ($\forall v \in (0, 1]$).

† DOUBLE-Q is no worse ($\forall v > \frac{1}{2}$) and better in low variations ($\forall v \leq \frac{1}{2}$).

‡ DOUBLE-Q is no worse ($\forall v \leq \frac{1}{2}$) and better in high variations ($\forall v > \frac{1}{2}$).

Other time-invariant COCO works adopted the hard violation metric, which prohibits compensation for constraint violations [17]-[19]. The saddle-point-type algorithm in [17] achieved $\mathcal{O}(\max\{T^c, T^{1-c}\})$ regret and $\mathcal{O}(T^{1-\frac{c}{2}})$ hard violation. Under Slater's condition, the violation was improved to $\mathcal{O}(T^{\frac{1-c}{2}})$ [18]. The lower-bounded virtual queue algorithm in [19] achieved $\mathcal{O}(1)$ hard violation without Slater's condition.

B. Time-Varying COCO

Recent COCO works considered unknown arbitrarily time-varying constraints under both soft [20], [21] and hard violation metrics [19], [22], [23]. The saddle-point-type algorithm in [20] achieved $\mathcal{O}(\sqrt{T})$ regret and $\mathcal{O}(T^{\frac{3}{4}})$ soft violation without Slater's condition, while the best-known $\mathcal{O}(\sqrt{T})$ soft violation was achieved by the virtual-queue-based algorithm in [21] under Slater's condition. The lower-bounded virtual queue algorithm in [19] achieved $\mathcal{O}(\sqrt{T})$ regret and $\mathcal{O}(T^{\frac{3}{4}})$ hard violation, while the best-known $\mathcal{O}(\sqrt{T}\log T)$ hard violation was achieved by the surrogate gradient descent algorithm in [22] and its optimistic version in [23] (under the standard COCO setting without predictions). However, these results reveal a significant $\Omega(\sqrt{T})$ performance gap in constraint violation between the two COCO worlds.

Another line of work bridges the two COCO worlds through unified analysis of constraint violations [24]-[27]. Specifically, the modified saddle-point-type algorithm in [24] achieved $\mathcal{O}(T^{\frac{1+v}{2}})$ regret and $\mathcal{O}(T^{1-v})$ soft violation under a sufficiently large Slater constant, where $v \in [0, 1]$ quantifies constraint variation (0: fixed constraints; 1: arbitrary constraints). Virtual-queue-based algorithms achieved $\mathcal{O}(\sqrt{T})$ regret and $\mathcal{O}(\max\{T^{\frac{3}{4}}, T^v\})$ soft violation without Slater's condition [25], improving to $\mathcal{O}(\max\{T^{\frac{1}{2}}, T^v\})$ soft violation under Slater's condition [26]. More recently, the doubly-bounded virtual queue algorithm in [27] achieved $\mathcal{O}(T^v)$ hard violation without Slater's condition — recovering $\mathcal{O}(1)$ violation for the first time in the time-varying COCO world. However, this hard violation bound provides no worst-case guarantees for arbitrary constraints, unlike those in [19], [22].

Addressing the significant performance gap between the two COCO worlds, we focus on centralized COCO. While distributed COCO has been studied [28]-[31], it maintains centralized performance bounds at best. As Table I demonstrates, DOUBLE-Q *bridges the best-known hard violations in the two COCO worlds, outperforming state-of-the-art algorithms.*

III. COCO

A. Problem Formulation

The COCO problem can be modeled as a repeated game between a learner and an adversary over T time slots. At each time $t = 1, \ldots, T$, the learner first commits a decision $x_t \in \mathcal{X}$, where $\mathcal{X} \subseteq \mathbb{R}^p$ is a known convex feasible set. The adversary — potentially having observed x_t — then reveals a convex loss function $f_t(x) : \mathcal{X} \to \mathbb{R}$ and N constraint functions $\{g_t^n(x) : \mathcal{X} \to \mathbb{R}\}_{n=1}^N$, both of which may change *arbitrarily* over time. Consequently, the learner incurs a loss $f_t(x_t)$ and constraint violations $\{g_t^n(x_t)\}_{n=1}^N$. Note that when selecting x_t, the learner only has *delayed* feedback information on the loss function $f_{t-1}(x)$ and constraint functions $\{g_{t-1}^n(x)\}_{n=1}^N$.

The learner aims to select a sequence of feasible decisions that minimizes the cumulative loss over the time horizon T, while satisfying all constraints at each time t. This *time-varying* COCO problem [19]-[27] is formulated as follows:

$$\mathbf{P}: \quad \min_{\{x_t \in \mathcal{X}\}_{t=1}^T} \quad \sum_{t=1}^T f_t(x_t)$$
$$\text{s.t.} \quad g_t^n(x_t) \le 0, \quad \forall t,\ \forall n, \tag{1}$$

which generalizes the *time-invariant* COCO problem [14]-[19] where $g_t^n(x) = g^n(x)$ holds for all t.

The tractability of problem **P** fundamentally depends on the cumulative temporal variation in constraints $\{g_t^n(x)\}_{t=1,n=1}^{T,N}$. To enable a unified constraint violation analysis for both time-varying and time-invariant COCO problems, we adopt the following cumulative *constraint variation* measure [24]-[27]:

$$\sum_{t=1}^{T-1} \sum_{n=1}^N \max_{x \in \mathcal{X}} \left| g_{t+1}^n(x) - g_t^n(x) \right| = \mathcal{O}(T^v), \tag{2}$$

where $v \in [0, 1]$ continuously quantifies the constraint variation from 0 for fixed constraints to 1 for arbitrary constraints.

B. Performance Metrics

Since $f_t(x)$ and $\{g_t^n(x)\}_{n=1}^N$ are unknown when selecting x_t, the optimal solution to **P** cannot be determined causally. The COCO framework therefore evaluates online algorithms by comparing their decisions $\{x_t\}_{t=1}^T$ against the best fixed solution $x^\star \in \arg\min_{x \in \mathcal{X}} \{\sum_{t=1}^T f_t(x) \big| g_t^n(x) \le 0, \forall t, \forall n\}$, which solves the *offline* counterpart of **P** with complete knowledge of all loss functions $\{f_t(x)\}_{t=1}^T$ and constraint functions $\{g_t^n(x)\}_{t=1,n=1}^{T,N}$ in hindsight. The performance gap between $\{x_t\}_{t=1}^T$ and $x^\star$ is quantified through the *regret* [19]-[27]:

$$\mathrm{REG}(T) \triangleq \sum_{t=1}^T [f_t(x_t) - f_t(x^\star)]. \tag{3}$$

Moreover, since $\{g_t^n(x)\}_{n=1}^N$ are unknown when selecting x_t, violations of constraints (1) may occur at each time. These constraint violations are quantified cumulatively through the *hard violation* [19], [22], [23], [27]:

$$\mathrm{VIO}(T) \triangleq \sum_{t=1}^T \sum_{n=1}^N [g_t^n(x_t)]_+, \tag{4}$$

where $[a]_+ = \max\{a, 0\}$. This violation metric is termed "hard" because it accumulates only positive constraint violations, prohibiting temporal compensation. Note that any bound on the hard violation readily holds for the *soft* violation $\sum_{n=1}^N [\sum_{t=1}^T g_t^n(x_t)]_+$ [20], [21], [24]-[26].

C. Assumptions

To bound the regret (3) and hard violation (4), we impose the following standard assumptions on the feasible set, loss functions, and constraint functions [19], [22], [27]:

Assumption 1. (Bounded Set) The feasible set is bounded, *i.e.,* $\exists R > 0$ such that $\|x - y\| \le R, \forall x, y \in \mathcal{X}$.

Assumption 2. (Bounded Gradients) The loss and constraint gradients are bounded, *i.e.,* $\exists D > 0$ such that $\|\nabla f_t(x)\| \le D$ and $\|\nabla g_t^n(x)\| \le D, \forall x \in \mathcal{X}, \forall t, \forall n$.

Assumption 3. (Bounded Constraints) The constraints are bounded, *i.e.,* $\exists G > 0$ such that $|g_t^n(x)| \le G, \forall x \in \mathcal{X}, \forall t, \forall n$.

IV. DOUBLE-Q

In this section, we present DOUBLE-Q, a novel COCO algorithm leveraging two virtual queues (original and surrogate) to minimize a surrogate drift-plus-penalty (SDPP). While similar queues have been *separately* designed [22], [27], a *double-queue* COCO algorithm under this *SDPP* framework is new.

A. Original Virtual Queue

To guarantee a bounded violation for arbitrary constraints, we introduce an *original* virtual queue Q_t^n initialized as $Q_0^n = 0$ for each constraint n. At each time $t \ge 1$, after the adversary reveals $g_t^n(x)$, the learner updates:

$$Q_t^n = Q_{t-1}^n + \beta [g_t^n(x_t)]_+, \tag{5}$$

where $\beta > 0$ is a scaling factor on the per-slot hard violation. This update rule (5) directly relates queue values to the hard violation (4) through the identity:

$$\mathrm{VIO}(T) = \frac{1}{\beta} \sum_{n=1}^N Q_T^n. \tag{6}$$

Similar to the drift-plus-penalty approach (DPP) for stochastic network optimization [32], we define an *original* Lyapunov drift for each $t \ge 1$ as $\Delta_t \triangleq \sum_{n=1}^N [\Phi(Q_t^n) - \Phi(Q_{t-1}^n)]$, where $\Phi(\cdot) : [0, +\infty) \to [0, +\infty)$ is a non-decreasing convex differentiable Lyapunov function with $\Phi(0) = 0$. The definitions of $\Phi(\cdot)$ depend on whether the losses are convex or strongly convex, as will be specified in Lemmas 2 and 3, respectively. The following proposition provides an upper bound on Δ_t.

Proposition 1. (Original Drift Bound) The original Lyapunov drift Δ_t is upper bounded for any $t \ge 1$ as:

$$\Delta_t \le \beta \sum_{n=1}^N \Phi'(Q_t^n) [g_t^n(x_t)]_+. \tag{7}$$

Proof: The proof follows directly from the convexity of $\Phi(\cdot)$ and the definition of Q_t^n in (5). ■

Based on the above original drift upper bound, we define a *surrogate* loss function $\hat{f}_t(x) : \mathcal{X} \to \mathbb{R}$ for each $t \geq 1$ as:

$$\hat{f}_t(x) \triangleq U f_t(x) + \beta \sum_{n=1}^{N} \Phi'(Q_t^n)[g_t^n(x)]_+, \tag{8}$$

where $U > 0$ is a weighting factor on the original loss function. Let $\Delta_t + U f_t(x)$ be the *original* DPP. From the original drift bound (7) and the definition of the surrogate loss function (8), we can see the intuition behind constructing $\hat{f}_t(x)$: *minimizing the surrogate loss is equivalent to minimizing an upper bound on the original DPP*[1]

$$\underbrace{\Delta_t + U f_t(x_t)}_{\text{original drift-plus-penalty}} \leq \hat{f}_t(x_t). \tag{9}$$

B. Surrogate Virtual Queue

To guarantee a minimum violation for fixed constraints, we introduce a *surrogate* virtual queue $\hat{Q}_t^n$ initialized as $\hat{Q}_1^n = V$ for each constraint n. At each time $t > 1$, after the adversary reveals $g_t^n(x)$, the learner updates:

$$\hat{Q}_t^n = \max\left\{(1-\epsilon)\hat{Q}_{t-1}^n + [g_t^n(x_t)]_+, V\right\}, \tag{10}$$

where $\epsilon \in (0,1)$ is a penalty factor on the previous queue and $V \in (0, \frac{G}{\epsilon})$ is a minimum queue length. The following proposition provides bounds on the surrogate virtual queue.

Proposition 2. (Surrogate Queue Bounds) Under Assumption 3, the surrogate virtual queue $\hat{Q}_t^n$ is bounded from both below and above for any $t \geq 1$ and n as:

$$V \leq \hat{Q}_t^n \leq \frac{G}{\epsilon}. \tag{11}$$

Proof: The proof follows from noting that (11) holds for $t = 1$ by initialization and then using the updating rule (10) to show (11) holds for any $t > 1$ by induction. ■

Unlike the original virtual queue Q_t^n, which directly connects to the hard violation $\mathrm{VIO}(T)$ via (6), bounds on the surrogate virtual queue $\hat{Q}_t^n$ in (11) cannot be directly translated to a bound on $\mathrm{VIO}(T)$. To establish this connection, we define a *surrogate* Lyapunov drift for each time $t > 1$ as $\hat{\Delta}_{t-1} \triangleq \frac{1}{2}\sum_{n=1}^{N}[(\hat{Q}_t^n - V)^2 - (\hat{Q}_{t-1}^n - V)^2]$. The following proposition provides an upper bound on the surrogate Lyapunov drift $\hat{\Delta}_{t-1}$, connecting $\hat{Q}_t^n$ and $\mathrm{VIO}(T)$.

Proposition 3. (Surrogate Drift Bound) Under Assumption 3, the surrogate Lyapunov drift $\hat{\Delta}_{t-1}$ is upper bounded for any $t > 1$ as:

$$\begin{aligned}\hat{\Delta}_{t-1} \leq& \sum_{n=1}^{N} \hat{Q}_{t-1}^n [g_{t-1}^n(x_t)]_+ - V \sum_{n=1}^{N} [g_t^n(x_t)]_+ \\ &+ \frac{G}{\epsilon}\sum_{n=1}^{N} \max_{x\in\mathcal{X}} |g_t^n(x) - g_{t-1}^n(x)| + 2NG^2. \end{aligned} \tag{12}$$

Proof: Substituting $\hat{Q}_t^n$ in (10) into $\hat{\Delta}_{t-1}$, we have $\hat{\Delta}_{t-1} \leq \sum_{n=1}^{N}[\hat{Q}_{t-1}^n[g_{t-1}^n(x_t)]_+ - V[g_t^n(x_t)]_+ + \hat{Q}_{t-1}^n([g_t^n(x_t)]_+ - [g_{t-1}^n(x_t)]_+) + \frac{1}{2}([g_t^n(x_t)]_+ - \epsilon\hat{Q}_{t-1}^n)^2 - \epsilon\hat{Q}_{t-1}^n(\hat{Q}_{t-1}^n - V)]$, which can then be bounded using the queue bounds in (11). ■

[1]This intuition is different from the regret decomposition in [22] by incorporating the original Lyapunov drift and the DPP framework.

Algorithm 1 DOUBLE-Q

Input: Non-decreasing learning rate $\{\alpha_t > 0\}_{t=1}^T$, scaling factor on per-slot hard violation $\beta > 0$, non-decreasing convex differentiable Lyapunov function $\Phi(\cdot) : [0, +\infty) \to [0, +\infty)$ with $\Phi(0) = 0$, weighting factor on original loss $U > 0$, minimum surrogate queue length $V \in (0, \frac{G}{\epsilon})$, and penalty factor on surrogate queue $\epsilon \in (0, 1)$.
Output: Decision sequence $\{x_t\}_{t=1}^T$.

```
Initialize x_1 ∈ X and Q_0^n = 0, Q̂_1^n = V, ∀n.
for each time t = 2, ..., T do
    Update x_t via solving P_t.
    Observe ∇f_t(x_t) and {g_t^n(x)}_{n=1}^N.
    Update Q_t^n, ∀n via (5).          ▷ original virtual Queue
    Compute ∇f̂_t(x_t) using (8).
    Update Q̂_t^n, ∀n via (10).        ▷ surrogate virtual Queue
end for
```

C. The DOUBLE-Q *Algorithm*

Algorithm: Using the surrogate loss $\hat{f}_t(x)$ in (8) and surrogate virtual queue $\hat{Q}_t^n$ in (10), we convert the original COCO problem $\mathbf{P}$ into a sequence of per-slot optimization problems $\{\mathbf{P}_t\}_{t=1}^T$, defined as follows:

$$\begin{aligned}\mathbf{P}_t: \quad \min_{x\in\mathcal{X}} \quad & \langle \nabla\hat{f}_{t-1}(x_{t-1}), x - x_{t-1}\rangle + \alpha_{t-1}\|x - x_{t-1}\|^2 \\ &+ \sum_{n=1}^{N} \hat{Q}_{t-1}^n [g_{t-1}^n(x)]_+,\end{aligned}$$

where $\alpha_{t-1} > 0$ is a non-decreasing learning rate.

In $\mathbf{P}_t$, we leverage the previous loss gradient $\nabla f_{t-1}(x_{t-1})$ from $\nabla\hat{f}_{t-1}(x_{t-1})$ to minimize the regret $\mathrm{REG}(T)$ in (3), and both the previous constraint gradient $\nabla g_{t-1}^n(x_{t-1})$ from $\nabla\hat{f}_{t-1}(x_{t-1})$ and the previous constraint function $g_{t-1}^n(x)$ — weighted by the original virtual queue Q_{t-1}^n and surrogate virtual queue $\hat{Q}_{t-1}^n$ — to control the hard violation $\mathrm{VIO}(T)$ in (4). We summarize the DOUBLE-Q algorithm in Algorithm 1.

Intuition: From the surrogate drift bound (12) derived in Proposition 3 and the above per-slot problem $\mathbf{P}_t$, we can see the intuition behind constructing $\mathbf{P}_t$: *solving* $\boldsymbol{P}_t$ *is equivalent to minimizing an upper bound on the surrogate DPP*[2]

$$\begin{aligned}&\underbrace{\hat{\Delta}_{t-1} + \langle\nabla\hat{f}_{t-1}(x_{t-1}), x_t - x_{t-1}\rangle + \alpha_{t-1}\|x_t - x_{t-1}\|^2}_{\text{surrogate drift-plus-penalty}} \\ &\leq U\langle\nabla f_{t-1}(x_{t-1}), x_t - x_{t-1}\rangle + \alpha_{t-1}\|x_t - x_{t-1}\|^2 \\ &\quad + \beta\sum_{n=1}^{N}\Phi'(Q_{t-1}^n)\langle\nabla[g_{t-1}^n(x_{t-1})]_+, x_t - x_{t-1}\rangle \\ &\quad + \sum_{n=1}^{N}\hat{Q}_{t-1}^n[g_{t-1}^n(x_t)]_+ + C,\end{aligned} \tag{13}$$

where we have substituted $\hat{f}_t(x)$ in (8) into $\mathbf{P}_t$, and combined the last two terms on the right-hand-side (RHS) of (12) as a constant $C \triangleq \frac{G}{\epsilon}\sum_{n=1}^{N}\max_{x\in\mathcal{X}}|g_t^n(x) - g_{t-1}^n(x)| + 2NG^2$ independent of the decision x_t (the second term on the RHS of (12) is non-positive, and thus has been trivially omitted).

[2]This intuition is different from the DPP minimization in [27] by incorporating the surrogate loss and double virtual queues.

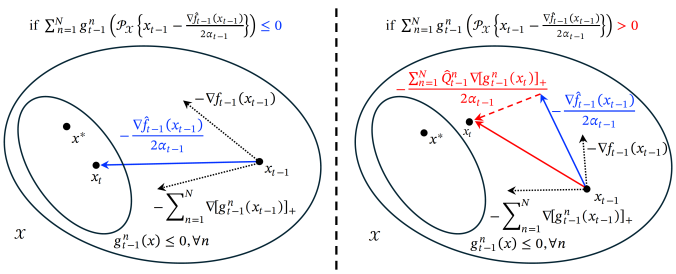

Fig. 1: An illustration of DOUBLE-Q's decision update: at each time t, the algorithm first performs projected surrogate gradient descent $x_t = \mathcal{P}_\mathcal{X}\{x_{t-1} - \frac{1}{2\alpha_{t-1}}\nabla \hat{f}_{t-1}(x_{t-1})\}$, then if this update incurs constraint violations, shifts x_t towards minimizing $\sum_{n=1}^N \hat{Q}_{t-1}^n[g_{t-1}^n(x)]_+$ to reduce hard violations.

Update: We derive the optimal solution x_t of $\mathbf{P}_t$. Minimizing the *surrogate penalty* alone in the objective of $\mathbf{P}_t$ over the feasible set $\mathcal{X}$ yields a *projected surrogate gradient descent* update $\mathcal{P}_\mathcal{X}\{x_{t-1} - \frac{1}{2\alpha_{t-1}}\nabla\hat{f}_{t-1}(x_{t-1})\}$, where $\mathcal{P}_\mathcal{X}\{x\} \triangleq \arg\min_{y\in\mathcal{X}} \|x-y\|^2$ is the projector onto $\mathcal{X}$. The additional minimization of the upper bound $\sum_{n=1}^N \hat{Q}_{t-1}^n[g_{t-1}^n(x)]_+ + C$ on *surrogate drift* $\hat{\Delta}_{t-1}$ adapts this update to hard violations. As illustrated in Fig. 1, the optimal x_t depends on whether hard violations occur during projected surrogate gradient descent, enabling our double-queue control of hard violations:

- If $\sum_{n=1}^N g_{t-1}^n(\mathcal{P}_\mathcal{X}\{x_{t-1} - \frac{1}{2\alpha_{t-1}}\nabla\hat{f}_{t-1}(x_{t-1})\}) \le 0$, *i.e.*, performing projected surrogate gradient descent incurs no constraint violations, the optimal solution to $\mathbf{P}_t$ is given by the surrogate gradient descent update:

$$x_t = \mathcal{P}_\mathcal{X}\Big\{x_{t-1} - \frac{U}{2\alpha_{t-1}}\nabla f_{t-1}(x_{t-1}) - \frac{\beta}{2\alpha_{t-1}}\sum_{n=1}^N \Phi'(Q_{t-1}^n)\nabla[g_{t-1}^n(x_{t-1})]_+\Big\}, \quad (14)$$

where we have substituted $\hat{f}_t(x)$ in (8) into $\nabla\hat{f}_{t-1}(x_{t-1})$. This update (14) dynamically balances gradients of the original loss $\nabla f_{t-1}(x_{t-1})$ and constraints $\{\nabla g_{t-1}^n(x_{t-1})\}_{n=1}^N$ through the original virtual queue Q_{t-1}^n, simultaneously minimizing loss while controlling hard violations. Compared with [27], the constraint gradient descent $\Phi'(Q_{t-1}^n)\nabla[g_{t-1}^n(x_{t-1})]_+$ provides additional robustness to arbitrary constraints in Theorem 2.

- If $\sum_{n=1}^N g_{t-1}^n(\mathcal{P}_\mathcal{X}\{x_{t-1} - \frac{1}{2\alpha_{t-1}}\nabla\hat{f}_{t-1}(x_{t-1})\}) > 0$, the last term $\sum_{n=1}^N \hat{Q}_{t-1}^n[g_{t-1}^n(x)]_+$ in the objective of $\mathbf{P}_t$ biases the surrogate gradient descent direction toward further reduction of constraint violations. Here, the surrogate virtual queue $\hat{Q}_{t-1}^n$ dynamically balances surrogate loss minimization and hard violation control. Compared with [19], the constraint penalty $\hat{Q}_{t-1}^n[g_{t-1}^n(x)]_+$ provides additional tightness to fixed constraints in Theorem 3.

Complexity: The computational complexity of DOUBLE-Q is dominated by solving $\mathbf{P}_t$. Due to its convexity, $\mathbf{P}_t$ can be solved efficiently in polynomial time.

V. PERFORMANCE BOUNDS OF DOUBLE-Q

We present new techniques to bound the regret and hard violation of DOUBLE-Q for both convex and strongly convex losses, particularly to account for both the original and surrogate virtual queues under the SDPP framework.

A. Bounds for Convex loss Functions

We first analyze the performance of DOUBLE-Q for convex loss functions under Assumptions 1-3. The following lemma provides a *per-slot* performance bound for DOUBLE-Q, derived from its minimization of the SDPP upper bound (13).

Lemma 1. (Per-Slot Bound for Convex Losses) Under Assumptions 1-3, for any $t < T$, DOUBLE-Q satisfies

$$U\big[f_t(x_t) - f_t(x^\star)\big] + \Delta_t + \sum_{n=1}^N \hat{Q}_t^n[g_t^n(x_{t+1})]_+ \le \alpha_t\big(\|x^\star - x_t\|^2 - \|x^\star - x_{t+1}\|^2\big) + \frac{\|\nabla\hat{f}_t(x_t)\|^2}{4\alpha_t}. \quad (15)$$

Proof: See Appendix A.

Building on Lemma 1, the following lemma provides a *cumulative* performance bound for DOUBLE-Q.

Lemma 2. (Cumulative Bound for Convex Losses) Under Assumptions 1-3, let $\alpha_t = \frac{1}{R}\sqrt{\frac{\sum_{\tau=1}^t \|\nabla\hat{f}_\tau(x_\tau)\|^2}{2}}$, $\beta = \frac{1}{2RD}$, $\Phi(\cdot) = \exp(\frac{\cdot}{2\sqrt{T}}) - 1$, and $U = 1$, DOUBLE-Q satisfies:

$$\sum_{t=1}^{T-1}[f_t(x_t) - f_t(x^\star)] + \frac{1}{2}\sum_{n=1}^N \exp\left(\frac{Q_{T-1}^n}{2\sqrt{T}}\right) + \sum_{n=1}^N\sum_{t=1}^{T-1}\hat{Q}_t^n[g_t^n(x_{t+1})]_+ \le 2RD\sqrt{T} + N. \quad (16)$$

Proof: See Appendix B.

We provide regret and hard violation bounds for DOUBLE-Q in Theorems 1-3, by *separately* bounding the first (original loss), second (original queue), and third (surrogate queue) term on the left-hand-side (LHS) of (16) in Lemma 2. Notably, the non-negativity of both queue lengths ($Q_{T-1}^n \ge 0, \hat{Q}_t^n \ge 0$) and hard violation ($[g_t^n(x_{t+1})]_+ \ge 0$) enables us to bound the regret via (16) in Theorem 1 without worrying about the two queues. More importantly, the SDPP framework allows the impacts of the two (original and surrogate) virtual queues to *coexist* in (16), *without* conflicting with each other when *separately* bounding the two hard violations in Theorems 2 and 3. This is in contrast to the single-queue analysis in [19] and [27] that can only provide one hard violation bound.

Theorem 1. (Regret Bound #1) Under Assumptions 1-3, use the same α_t, β, $\Phi(\cdot)$, and U as in Lemma 2, the regret of DOUBLE-Q is upper bounded by:

$$\mathrm{REG}(T) \le RD(2\sqrt{T} + 1) + N. \quad (17)$$

Proof: See Appendix C.

The following theorem provides the first hard violation bound for DOUBLE-Q by bounding the original queue Q_{T-1}^n in (16) and then translating this bound to the hard violation $\mathrm{VIO}(T)$ via (6). Notably, the non-negativity of the surrogate queues ($\hat{Q}_t^n \ge 0$) and hard violations ($[g_t^n(x_{t+1})]_+$) allows us to bound the hard violation without worrying about the surrogate queues.

Theorem 2. (Hard Violation Bound #1) Under Assumptions 1-3, use the same α_t, β, $\Phi(\cdot)$, and U as in Theorem 1, the hard violation of DOUBLE-Q is upper bounded by:

$$\mathrm{VIO}(T) \leq 2N\sqrt{T}\ln\big(2RD(T+2\sqrt{T})+2N\big)+NG. \quad (18)$$

Proof: See Appendix D.

The following theorem provides the second hard violation bound for DOUBLE-Q, by bounding the surrogate-queue-weighted violation $\hat{Q}_{t-1}^n[g_{t-1}^n(x_t)]_+$ in (16), and then translating to $\mathrm{VIO}(T)$ via (12). Notably, the non-negativity of the original queue Q_{T-1}^n allows us to bound the hard violation again without worrying about the original queues.

Theorem 3. (Hard Violation Bound #2) Under Assumptions 1-3, use the same α_t, β, $\Phi(\cdot)$, U as in Theorems 1 and 2, and let $V=\frac{GT}{2}$ and $\epsilon=\frac{1}{T}$, the hard violation of DOUBLE-Q is also upper bounded by:

$$\begin{aligned}\mathrm{VIO}(T) \leq & 2\sum_{t=1}^{T-1}\sum_{n=1}^{N}\max_{x\in\mathcal{X}}|g_{t+1}^n(x)-g_t^n(x)|+\frac{4RD}{G\sqrt{T}} \\ &+\frac{4NG}{T}+\frac{2N}{GT}+\frac{2RD}{G}+NG. \end{aligned}\quad (19)$$

Proof: See Appendix E.

B. Bounds for Strongly Convex Loss Functions

We further analyze the performance of DOUBLE-Q for strongly convex loss functions under Assumptions 1-4.

Assumption 4. (Strong Convexity) The loss functions are 2μ-strongly convex, *i.e.,* $\exists\mu>0$ such that $f_t(y)\geq f_t(x)+\langle\nabla f_t(x),y-x\rangle+\mu\|y-x\|^2,\forall x,y\in\mathcal{X},\forall t$.

We leverage strong convexity to refine the per-slot bound in Lemma 1 and then sum these improvements over time to provide a cumulative performance bound for DOUBLE-Q.

Lemma 3. (Cumulative Bound for Strongly Convex Losses) Under Assumptions 1-4, let $\alpha_t=U\mu t$, $\beta=1$, $\Phi(\cdot)=(\cdot)^2$, and $U=\frac{4ND^2\ln(Te)}{\mu}$, DOUBLE-Q satisfies:

$$\begin{aligned}&\frac{4ND^2\ln(Te)}{\mu}\sum_{t=1}^{T-1}\big[f_t(x_t)-f_t(x^\star)\big]+\frac{1}{2N}\Big(\sum_{n=1}^{N}Q_{T-1}^n\Big)^2 \\ &+\sum_{n=1}^{N}\sum_{t=1}^{T-1}\hat{Q}_t^n[g_t^n(x_{t+1})]_+\leq\frac{2ND^4[\ln(Te)]^2}{\mu^2}. \end{aligned}\quad (20)$$

Proof: See Appendix F.

Lemma 3 is the key to bound the regret and hard violation of DOUBLE-Q under strongly convex losses. Leveraging the double-queue design under the SDPP framework, we provide a regret bound and two hard violation bounds for strongly convex losses in Theorems 4-6.

Theorem 4. (Regret Bound #2) Under Assumptions 1-4, use the same α_t, β, $\Phi(\cdot)$, and U as in Lemma 3, the regret of DOUBLE-Q is upper bounded by:

$$\mathrm{REG}(T)\leq\frac{D^2}{2\mu}\ln(Te)+RD. \quad (21)$$

Proof: See Appendix G.

Theorem 5. (Hard Violation Bound #3) Under Assumptions 1-4, use the same α_t, β, $\Phi(\cdot)$, and U as in Theorem 4, the hard violation of DOUBLE-Q is upper bounded by:

$$\mathrm{VIO}(T)\leq\frac{2\sqrt{2}ND}{\sqrt{\mu}}\sqrt{T\ln(Te)}+\frac{2ND^2}{\mu}\ln(Te)+NG. \quad (22)$$

Proof: See Appendix H.

Theorem 6. (Hard Violation Bound #4) Under Assumptions 1-4, use the same α_t, β, $\Phi(\cdot)$, and U as in Theorems 4 and 5, and let $V=\frac{GT\log T}{2}$ and $\epsilon=\frac{1}{T\log T}$, the hard violation of DOUBLE-Q is also upper bounded by:

$$\begin{aligned}\mathrm{VIO}(T)\leq & 2\sum_{t=1}^{T-1}\sum_{n=1}^{N}\max_{x\in\mathcal{X}}|g_{t+1}^n(x)-g_t^n(x)|+\frac{8ND^2\ln(Te)}{\mu G\log T} \\ &+\frac{4ND^4[\ln(Te)]^2}{\mu^2 GT\log T}+\frac{4NG^2}{GT\log T}. \end{aligned}\quad (23)$$

Proof: See Appendix I.

C. Regret and Hard Violation Bounds

Theorems 1-6 yield the following corollaries on the regret and hard violation bounds for DOUBLE-Q under convex and strongly convex losses.

Corollary 1. (Bounds for Convex Losses) Under Assumptions 1-3, let $\alpha_t=\frac{1}{R}\sqrt{\frac{\sum_{\tau=1}^t\|\nabla\hat{f}_\tau(x_\tau)\|^2}{2}}$, $\beta=\frac{1}{2RD}$, $\Phi(\cdot)=\exp(\frac{\cdot}{2\sqrt{T}})-1$, $U=1$, $V=\frac{GT}{2}$, and $\epsilon=\frac{1}{T}$, for any $v\in[0,1]$, DOUBLE-Q achieves:

$$\mathrm{REG}(T)=\mathcal{O}\big(\sqrt{T}\big), \quad (24)$$

$$\mathrm{VIO}(T)=\mathcal{O}\big(\min\big\{\sqrt{T}\log T,T^v\big\}\big), \quad (25)$$

where $\min$ comes from combining Theorems 2 and 3.

Corollary 1 shows that DOUBLE-Q *simultaneously* guarantees the best-known $\mathcal{O}(\sqrt{T}\log T)$ hard violation for arbitrary constraints at worst ($v=1$) and recovers the best-known $\mathcal{O}(1)$ hard violation for fixed constraints at best ($v=0$), without sacrificing the $\mathcal{O}(\sqrt{T})$ regret. Notably, as Table I demonstrates, DOUBLE-Q is *never worse* and can be *strictly better* than state-of-the-art COCO algorithms [19], [22], [27] for *any* constraint variation ($\forall v\in[0,1]$).

Corollary 2. (Bounds for Strongly Convex Losses) Under Assumptions 1-4, let $\alpha_t=U\mu t$, $\beta=1$, $\Phi(\cdot)=(\cdot)^2$, $U=\frac{4ND^2\ln(Te)}{\mu}$, $V=\frac{GT\log T}{2}$, and $\epsilon=\frac{1}{T\log T}$, for any $v\in[0,1]$, DOUBLE-Q achieves:

$$\mathrm{REG}(T)=\mathcal{O}\big(\log T\big), \quad (26)$$

$$\mathrm{VIO}(T)=\mathcal{O}\big(\min\big\{\sqrt{T\log T},T^v\big\}\big). \quad (27)$$

where $\min$ comes from combining Theorems 5 and 6.

Corollary 2 shows that for strongly convex loss functions, DOUBLE-Q simultaneously guarantees the best-known $\mathcal{O}(\sqrt{T\log T})$ hard violation at worst and recovers the best-known $\mathcal{O}(1)$ hard violation at best, without sacrificing the $\mathcal{O}(\log T)$ regret. Notably, DOUBLE-Q is still never worse and can be strictly better than state-of-the-art COCO algorithms [19], [22], [27] for any constraint variation.

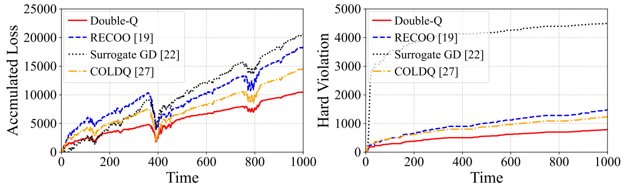

Fig. 2: Online quadratic programming with time-varying constraints.

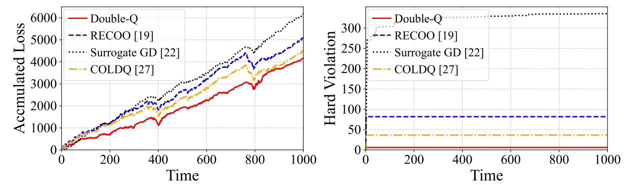

Fig. 3: Online quadratic programming with time-invariant constraints.

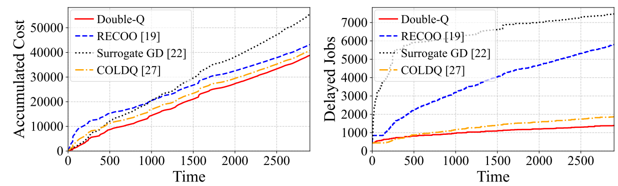

Fig. 4: Online job scheduling.

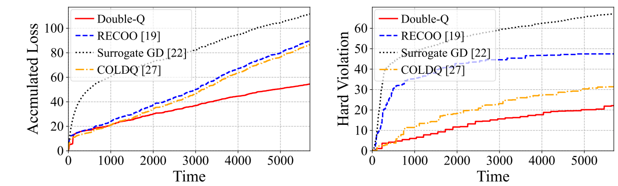

Fig. 5: Online fraud detection.

VI. Simulation Results

In this section, we conduct numerical experiments to compare Double-Q with the state-of-the-art COCO algorithms: i) RECtified Online Optimization (RECOO) [19] which uses a lower-bounded virtual queue, ii) Surrogate Gradient Descent (Surrogate GD) [22] which uses a standard virtual queue to construct a surrogate loss, and iii) Constrained Online Learning with Doubly-bounded Queue (COLDQ) [27] which uses a lower-and-upper-bounded virtual queue. We adopt the suggested algorithm parameter settings in [19], [22], [27] and fine-tune the constant parameters that do not impact the theoretical convergence rates to reach their best performance.

A. Online Quadratic Programming

We first apply Double-Q to online quadratic programming under both time-varying and time-invariant constraints as in [19], [27]. We set the loss as $f_t(x) = \|x - \theta_t\|^2 + 40\langle\theta_t, x\rangle$, where $\theta_t = \theta_t^1 + \theta_t^2 + \theta_t^3 \in \mathbb{R}^2$. Here, $\theta_t^1[i] \sim \mathrm{U}(-t^{1/10}, t^{1/10}), \forall i$ is a slow-drift term, $\theta_t^2[i] \sim \mathrm{U}(-1,0), \forall i$ for $t \in [1,350] \cup [400,750] \cup [800,1000]$ and $\theta_t^2[i] \sim \mathrm{U}(0,1), \forall i$ otherwise is a piecewise-bias term, and $\theta_t^3[i] = (-1)^{\mu_t}, \forall i$ is a binary-flip term with $\{\mu_t\}_{t=1}^{1000}$ being a random permutation of 1 to 1000. We set the time-varying constraints as $g_t(x) = A_t x - b_t$, where $A_t \in \mathbb{R}^{3\times 2}$ with $A_t[i,j] \sim \mathrm{U}(5,25), \forall i,j$ and $b_t \in \mathbb{R}^3$ with $b_t[i] \sim \mathrm{U}(0,10), \forall i$. We set the time-invariant constraints as $g(x) = Ax - b$ with $A[i,j] \in \mathrm{U}(0,60), \forall i,j$ and $b[i] \in \mathrm{U}(0,20), \forall i$, and the feasible set as $\mathcal{X} = \{-1 \le x[i] \le 1, \forall i\}$. We present the averaged results over 20 runs. Figs. 2 and 3 show that Double-Q achieves the lowest accumulated loss and over 30% lower hard violation compared to the other baselines.

B. Online Job Scheduling

We then apply Double-Q to online job scheduling as in [19], [27]. The application aims to reduce the energy cost while satisfying quality-of-service constraints. We use real-world electricity price data [33] over 10 metropolitan regions, each hosting 10 server clusters. The job scheduler allocates power $x \in \mathbb{R}^{100}$ to the clusters over a feasible set $\mathcal{X} = \{0 \le x[i] \le 100, \forall i\}$ at each time slot t of 5 minutes. The energy cost is $f_t(x) = \langle c_t, x\rangle$, where $c_t \in \mathbb{R}^{100}$ denotes the electricity prices. The quality-of-service constraint is $g_t(x) = \lambda_t - \sum_{i=1}^{100} h(x[i])$, where λ_t is the number of arriving jobs following a Poisson distribution with mean 400, and $h(x[i]) = 4\log(1 + 4x[i])$ denotes the service capacity of cluster i. Fig. 4 shows that Double-Q achieves the lowest energy cost without sacrificing any service quality, compared with RECOO, Surrogate GD, and COLDQ.

C. Online Fraud Detection

Finally, we apply Double-Q to online fraud detection using non-convex neural networks as in [22]. We use a real-world credit card transaction dataset [34]. We train a single-layer neural network with weights $x \in \mathbb{R}^{321}$ to predict the label of each transaction before its true label is revealed. We adopt the log-likelihood loss $f_t(x) = \frac{1}{50}\sum_{i=1}^{50} y_t^i \log(\hat{y}_t^i(z_t^i, x)) + (1 - y_t^i)\log(1 - \hat{y}_t^i(z_t^i, x))$ over a mini-batch of 50 data samples at each time t, where $(z_t^i \in \mathbb{R}^{30}, y_t^i \in \{0,1\})$ and $\hat{y}_t^i \in (0,1)$ denote the i-th data sample and its probability estimate yielded by the neural network. Similar to the constrained online binary classification problem considered in [15], we adopt the time-varying mixed-norm constraint $\|x\|_1 + \frac{1}{2}\|x\|^2 \le \rho_t$, where $\rho_t = 8\mathrm{U}(0,1) + 40$. Fig. 5 shows that Double-Q substantially outperforms Surrogate GD and achieves over 30% lower accumulated training loss compared to RECOO and COLDQ, while maintaining the lowest hard violation.

VII. Conclusions

We propose Double-Q, an effective algorithm for COCO in both worlds of time-varying and time-invariant constraints. Our novel original-and-surrogate double-queue design enables tighter dual violation control compared to existing single-queue approaches. Through a new surrogate drift-plus-penalty analysis, Double-Q achieves $\mathcal{O}(\sqrt{T})$ regret and $\mathcal{O}(\min\{\sqrt{T}\log T, T^v\})$ hard violation. This violation bound is the first to bridge the best-known results across the two COCO worlds, guaranteeing $\mathcal{O}(\sqrt{T}\log T)$ hard violation for arbitrary constraints while recovering $\mathcal{O}(1)$ hard violation for fixed constraints. For strongly convex loss functions, Double-Q further achieves $\mathcal{O}(\log T)$ regret and $\mathcal{O}(\min\{\sqrt{T\log T}, T^v\})$ hard violation. Simulation results demonstrate the superior performance of Double-Q over state-of-the-art baselines.

The authors have provided public access to their code and data at https://github.com/weiyi-qin/INFOCOM2026-DoubleQ.

Appendix A: Proof of Lemma 1

Proof: The objective of $\mathbf{P}_t$ is $2\alpha_{t-1}$-strongly convex due to $\alpha_{t-1}\|x-x_{t-1}\|^2$. From the optimality condition of strongly convex function (Lemma 2.8 in [1]), we have for any $t>1$

$$\begin{aligned}&\langle \nabla \hat{f}_{t-1}(x_{t-1}), x_t - x_{t-1}\rangle + \alpha_{t-1}\|x_t - x_{t-1}\|^2\\&\quad+\sum_{n=1}^{N}\hat{Q}_{t-1}^n[g_{t-1}^n(x_t)]_+\\&\le \langle \nabla \hat{f}_{t-1}(x_{t-1}), x^\star - x_{t-1}\rangle + \alpha_{t-1}\|x^\star - x_{t-1}\|^2\\&\quad+\sum_{n=1}^{N}\hat{Q}_{t-1}^n[g_{t-1}^n(x^\star)]_+ - \alpha_{t-1}\|x^\star - x_t\|^2. \qquad (28)\end{aligned}$$

Rearranging terms of (28) and noting that $g_{t-1}^n(x^\star)\le 0$ from the definition of $x^\star$ below (3), we have

$$\begin{aligned}&\sum_{n=1}^{N}\hat{Q}_{t-1}^n[g_{t-1}^n(x_t)]_+ \le \langle \nabla \hat{f}_{t-1}(x_{t-1}), x^\star - x_{t-1}\rangle\\&\quad+\alpha_{t-1}\big(\|x^\star - x_{t-1}\|^2 - \|x^\star - x_t\|^2\big)\\&\quad+\langle \nabla \hat{f}_{t-1}(x_{t-1}), x_{t-1} - x_t\rangle - \alpha_{t-1}\|x_t - x_{t-1}\|^2. \qquad (29)\end{aligned}$$

Due to the convexity of the surrogate loss function $\hat{f}_t(x)$, the first term on the RHS of (29) can be bounded as

$$\begin{aligned}&\langle \nabla \hat{f}_{t-1}(x_{t-1}), x^\star - x_{t-1}\rangle \le \hat{f}_{t-1}(x^\star) - \hat{f}_{t-1}(x_{t-1})\\&\quad\le U\big[f_{t-1}(x^\star) - f_{t-1}(x_{t-1})\big] - \Delta_{t-1}, \qquad (30)\end{aligned}$$

where the last inequality follows from applying the original DPP upper bound in (9) and noting that $\hat{f}_{t-1}(x^\star) = Uf_{t-1}(x^\star) + \beta\sum_{n=1}^{N}\nabla\Phi(Q_{t-1}^n)[g_{t-1}^n(x^\star)]_+ = Uf_{t-1}(x^\star)$.

Substituting (30) into (29), applying $2\langle a,b\rangle \le \|a\|^2 + \|b\|^2$ to bound the last two terms on the RHS of (29) as $\frac{1}{4\alpha_{t-1}}\|\nabla \hat{f}_{t-1}(x_{t-1})\|^2$, and rearranging terms, we have (15). ∎

Appendix B: Proof of Lemma 2

Proof: To facilitate our proof, we let $D_t = \|\nabla \hat{f}_t(x_t)\|$ and $S_t = \sum_{\tau=1}^{t} D_\tau^2$. Summing (15) over $t=1,\dots,T-1$, yields

$$\begin{aligned}&U\sum_{t=1}^{T-1}\big[f_t(x_t) - f_t(x^\star)\big] + \sum_{t=1}^{T-1}\Delta_t + \sum_{n=1}^{N}\sum_{t=1}^{T-1}\hat{Q}_t^n[g_t^n(x_{t+1})]_+\\&\le \alpha_1\|x^\star - x_1\|^2 + \sum_{t=2}^{T-1}\|x^\star - x_t\|^2(\alpha_t - \alpha_{t-1}) + \sum_{t=1}^{T-1}\frac{D_t^2}{4\alpha_t}\\&\overset{(a)}{\le} R^2\alpha_{T-1} + \sum_{t=1}^{T-1}\frac{D_t^2}{4\alpha_t} \overset{(b)}{\le} R\sqrt{2S_{T-1}}, \qquad (31)\end{aligned}$$

where (a) is because $\mathcal{X}$ being bounded in Assumption 1, and (b) follows from choosing $\alpha_t = \frac{1}{R}\sqrt{\frac{S_t}{2}}$ and applying $\frac{D_t^2}{\sqrt{S_t}} = \int_{S_{t-1}}^{S_t}\frac{1}{\sqrt{S_t}}dx \le \int_{S_{t-1}}^{S_t}\frac{1}{\sqrt{x}}dx$ such that $R^2\alpha_{T-1}$ and $\sum_{t=1}^{T-1}\frac{D_t^2}{4\alpha_t}$ share the same upper bound as $R^2\alpha_{T-1} = R\sqrt{\frac{S_{T-1}}{2}}$ and

$$\sum_{t=1}^{T-1}\frac{D_t^2}{4\alpha_t} = \frac{R}{2\sqrt{2}}\sum_{t=1}^{T-1}\frac{D_t^2}{\sqrt{S_t}} \le \frac{R}{2\sqrt{2}}\int_0^{S_{t-1}}\frac{1}{\sqrt{x}}dx = R\sqrt{\frac{S_{T-1}}{2}}.$$

We now bound the RHS of (31). We have

$$S_{T-1} \le T\Big[UD + D\beta\sum_{n=1}^{N}\Phi'(Q_{T-1}^n)\Big]^2, \qquad (32)$$

which follows from the definition of $\hat{f}_t(x)$ in (8), the triangle inequality, and $\Phi'(x)$ and Q_t^n being non-decreasing, such that $D_t = \|\nabla \hat{f}_t(x_t)\| \le U\|\nabla f_t(x_t)\| + \beta\sum_{n=1}^{N}\Phi'(Q_t^n)\|\nabla g_t^n(x_t)\| \le UD + \beta D\sum_{n=1}^{N}\Phi'(Q_{T-1}^n)$.

Substituting (32) into (31) and applying $\sqrt{2(a+b)^2}\le 2a + 2b, \forall a,b\ge 0$, noting that $\sum_{t=1}^{T-1}\Delta_t = \sum_{n=1}^{N}[\Phi(Q_{T-1}^n) - \Phi(Q_0^n)] = \sum_{n=1}^{N}\Phi(Q_{T-1}^n)$, and rearranging terms, we have

$$\begin{aligned}&U\sum_{t=1}^{T-1}\big[f_t(x_t) - f_t(x^\star)\big] + \sum_{n=1}^{N}\sum_{t=1}^{T-1}\hat{Q}_t^n[g_t^n(x_{t+1})]_+\\&\le 2URD\sqrt{T} + \sum_{n=1}^{N}\big[2\beta RD\sqrt{T}\Phi'(Q_{T-1}^n) - \Phi(Q_{T-1}^n)\big]. \quad (33)\end{aligned}$$

From (33), we can see the intuition behind choosing $\beta = \frac{1}{2RD}$ and $\Phi(x) = \exp(\frac{x}{2\sqrt{T}}) - 1$ with $\Phi'(x) = \frac{1}{2\sqrt{T}}\exp(\frac{x}{2\sqrt{T}})$, which aims to bound the last term on the RHS of (33) by a constant via

$$2\beta RD\sqrt{T}\Phi'(Q_{T-1}^n) - \Phi(Q_{T-1}^n) = -\frac{1}{2}\exp\Big(\frac{Q_{T-1}^n}{2\sqrt{T}}\Big) + 1.$$

Substituting the above inequality into (33), and let $U=1$, we have (16). ∎

Appendix C: Proof of Theorem 1

Proof: The proof follows from substituting $\exp(\frac{Q_{T-1}^n}{2\sqrt{T}})\ge 0, \forall n$ and $\hat{Q}_t^n[g_t^n(x_{t+1})]_+ \ge 0, \forall t<T, \forall n$ into (16), and further noting that $f_T(x_T) - f_T(x^\star) \le \langle \nabla f_T(x^\star), x^\star - x_T\rangle \le RD$ from the convexity of $f_T(x)$ and Assumptions 1 and 2. ∎

Appendix D: Proof of Theorem 2

Proof: Substituting $f_t(x^\star) - f_t(x_t) \le RD, \forall t$ and $\hat{Q}_t^n[g_t^n(x_{t+1})]_+ \ge 0, \forall t<T, \forall n$ into (16) of Lemma 2 yields

$$\frac{1}{2}\sum_{n=1}^{N}\exp\Big(\frac{Q_{T-1}^n}{2\sqrt{T}}\Big) \le RD(T + 2\sqrt{T}) + N. \qquad (34)$$

Based on (34), we now derive an upper bound on $\sum_{n=1}^{N}Q_{T-1}^n$. Since $\exp(\cdot)$ is convex, by Jensen's inequality, we have $\exp(\frac{1}{N}\sum_{n=1}^{N}\frac{Q_{T-1}^n}{2\sqrt{T}}) \le \frac{1}{N}\sum_{n=1}^{N}\exp(\frac{Q_{T-1}^n}{2\sqrt{T}})$. Taking the logarithm on both sides of the above inequality and multiplying both sides by $2N\sqrt{T}$ yields

$$\begin{aligned}\sum_{n=1}^{N}Q_{T-1}^n &\le 2N\sqrt{T}\ln\Big(\frac{1}{N}\sum_{n=1}^{N}\exp\Big(\frac{Q_{T-1}^n}{2\sqrt{T}}\Big)\Big)\\&\overset{(a)}{\le} 2N\sqrt{T}\ln\Big(\frac{2RD(T+2\sqrt{T})}{N} + 2\Big), \qquad (35)\end{aligned}$$

where (a) follows from substituting (34) and noting that $\ln(\cdot)$ is non-decreasing.

Further noting that $\mathrm{VIO}(T) = \frac{1}{\beta}\sum_{n=1}^{N}Q_T^n$ in (6), $Q_T^n = Q_{T-1}^n + \beta[g_T^n(x_T)]_+ \le Q_{T-1}^n + \beta G$ from the original virtual queue updating rule (5) and Assumption 3, and $\ln(\frac{a}{N}) \le \ln(a), \forall a>0, N\ge 1$, we have (18). ∎

Appendix E: Proof of Theorem 3

Substituting $U = 1$, $f_t(x^\star) - f_t(x_t) \le RD, \forall t$ and $\exp(\frac{Q^n_{T-1}}{2\sqrt{T}}) \ge 0, \forall n$ into (16) of Lemma 2 yields

$$\sum_{n=1}^{N}\sum_{t=1}^{T-1} \hat{Q}^n_t [g^n_t(x_{t+1})]_+ \le RD(T + 2\sqrt{T}) + N. \tag{36}$$

Summing (12) over $t = 2, \dots, T$, we have

$$\begin{aligned}\sum_{t=1}^{T-1} \hat{\Delta}_t \le& \sum_{t=1}^{T-1}\sum_{n=1}^{N} \hat{Q}^n_t [g^n_t(x_{t+1})]_+ - V\sum_{t=2}^{T}\sum_{n=1}^{N}[g^n_t(x_t)]_+ \\ &+ \frac{G}{\epsilon}\sum_{t=1}^{T-1}\sum_{n=1}^{N}\max_{x\in\mathcal{X}} |g^n_{t+1}(x) - g^n_t(x)| + 2NG^2. \end{aligned}\tag{37}$$

Substituting (37) into (36), dividing both sides by V, and rearranging terms, we have

$$\begin{aligned}\sum_{t=2}^{T}\sum_{n=1}^{N}[g^n_t(x_t)]_+ \le& \frac{G}{V\epsilon}\sum_{t=1}^{T-1}\sum_{n=1}^{N}\max_{x\in\mathcal{X}} |g^n_{t+1}(x) - g^n_t(x)| \\ &- \frac{1}{V}\sum_{t=1}^{T-1}\hat{\Delta}_t + \Big[\frac{2RD\sqrt{T}}{V} + \frac{N}{V} + \frac{2NG^2}{V} + \frac{RDT}{V}\Big]. \end{aligned}\tag{38}$$

We now bound the second term on the RHS of (38). From the design of the surrogate Lyapunov drift $\hat{\Delta}_t = \frac{1}{2}\sum_{n=1}^{N}[(\hat{Q}^n_{t+1} - V)^2 - (\hat{Q}^n_t - V)^2]$, $\hat{Q}^n_1 = V$ by initialization, and $\hat{Q}^n_t \ge V, \forall t$ in (6) of Proposition 2, we have $-\sum_{t=1}^{T-1}\hat{\Delta}_t = \frac{1}{2}\sum_{n=1}^{N}\big[(\hat{Q}^n_1 - V)^2 - (\hat{Q}^n_T - V)^2\big] \le 0$.

Substituting $-\sum_{t=1}^{T-1}\hat{\Delta}_t \le 0$ into (38), letting $V = \frac{GT}{2}$, $\epsilon = \frac{1}{T}$ to bound the last term on the RHS of (38) by $\mathcal{O}(1)$, and using $[g^n_1(x_1)]_+ \le G$ from Assumption 3, yields (19). ■

Appendix F: Proof of Lemma 3

Proof: The surrogate loss function $\hat{f}_{t-1}(x)$ is $2U\mu$-strongly convex due to $f_{t-1}(x)$ being 2μ-strongly convex in Assumption 4, which implies $\langle \nabla\hat{f}_{t-1}(x_{t-1}), x^\star - x_{t-1}\rangle \le \hat{f}_{t-1}(x^\star) - \hat{f}_{t-1}(x_{t-1}) - U\mu\|x^\star - x_{t-1}\|^2$. Substituting the above inequality into the RHS of (29) in the proof of Lemma 1 and following the rest of the proof, (15) becomes

$$\begin{aligned}&U\big[f_t(x_t) - f_t(x^\star)\big] + \Delta_t + \sum_{n=1}^{N}\hat{Q}^n_t[g^n_t(x_{t+1})]_+ \\ &\le (\alpha_t - U\mu)\|x^\star - x_t\|^2 - \alpha_t\|x^\star - x_{t+1}\|^2 + \frac{\|\nabla\hat{f}_t(x_t)\|^2}{4\alpha_t}.\end{aligned}$$

Summing the above inequality over $t = 1, \dots, T-1$ and noting that $\mathcal{X}$ being bounded in Assumption 1, we have

$$\begin{aligned}&U\sum_{t=1}^{T-1}\big[f_t(x_t) - f_t(x^\star)\big] + \sum_{t=1}^{T-1}\Delta_t + \sum_{n=1}^{N}\sum_{t=1}^{T-1}\hat{Q}^n_t[g^n_t(x_{t+1})]_+ \\ &\le (\alpha_1 - U\mu)R^2 + R^2\sum_{t=2}^{T-1}(\alpha_t - \alpha_{t-1} - U\mu) + \sum_{t=1}^{T-1}\frac{\|\nabla\hat{f}_t(x_t)\|^2}{4\alpha_t} \\ &\overset{(a)}{\le} \sum_{t=1}^{T-1}\frac{\|\nabla\hat{f}_t(x_t)\|^2}{4U\mu t} \overset{(b)}{\le} \sum_{t=1}^{T-1}\frac{U^2D^2 + \beta^2D^2\big[\sum_{n=1}^{N}\Phi'(Q^n_{T-1})\big]^2}{2U\mu t},\end{aligned}$$

where (a) follows from choosing $\alpha_t = U\mu t$ such that $\alpha_1 - U\mu$ and $\alpha_t - \alpha_{t-1} - U\mu$ are both 0, (b) is due to $\|\nabla\hat{f}_t(x_t)\| \le UD + \beta D\sum_{n=1}^{N}\Phi'(Q^n_{T-1})$ as proved below (32) and $(a+b)^2 \le 2a^2 + 2b^2$, and (c) is because the harmonic number satisfies $\sum_{t=1}^{T-1}\frac{1}{t} \le \ln(T-1) + 1 \le \ln(Te)$.

Substituting $\sum_{t=1}^{T-1}\Delta_t = \sum_{n=1}^{N}\Phi(Q^n_{T-1})$ and $\sum_{t=1}^{T-1}\frac{1}{t} \le \ln(Te)$ into the LHS and RHS of the above inequality, respectively, and rearranging terms, we have

$$\begin{aligned}&U\sum_{t=1}^{T-1}\big[f_t(x_t) - f_t(x^\star)\big] + \sum_{n=1}^{N}\sum_{t=1}^{T-1}\hat{Q}^n_t[g^n_t(x_{t+1})]_+ \\ &\le \frac{D^2\ln(Te)}{2\mu}\bigg(U + \frac{\beta^2\big[\sum_{n=1}^{N}\Phi'(Q^n_{T-1})\big]^2}{U}\bigg) - \sum_{n=1}^{N}\Phi(Q^n_{T-1}) \\ &\overset{(a)}{\le} \frac{UD^2\ln(Te)}{2\mu} + \bigg[\frac{2D^2\ln(Te)}{U\mu} - \frac{1}{N}\bigg]\Big(\sum_{n=1}^{N}Q^n_{T-1}\Big)^2,\end{aligned}\tag{39}$$

where (a) follows from $\beta = 1$, $\Phi(\cdot) = (\cdot)^2$, and the Cauchy-Schwarz inequality, such that both $[\sum_{n=1}^{N}\Phi'(Q^n_{T-1})]^2$ and $\sum_{n=1}^{N}\Phi(Q^n_{T-1})$ are converted to the same $(\sum_{n=1}^{N}Q^n_{T-1})^2$.

Choosing $U = \frac{4ND^2\ln(Te)}{\mu}$ such that the last term on the RHS of (39) is bound by 0, we have (20). ■

Appendix G: Proof of Theorem 4

Proof: Substituting $Q^n_{T-1} \ge 0, \forall n$ and $\hat{Q}^n_t[g^n_t(x_{t+1})]_+ \ge 0, \forall t < T, \forall n$ into (20), dividing both sides by $\frac{4ND^2\ln(Te)}{\mu}$, and noting that $f_T(x_T) - f_T(x^\star) \le RD$, we have (21). ■

Appendix H: Proof of Theorem 5

Proof: Substituting $f_t(x^\star) - f_t(x_t) \le RD, \forall t$ and $\hat{Q}^n_t[g^n_t(x_{t+1})]_+ \ge 0, \forall t < T, \forall n$ into (20), multiplying both sides by $2N$, and then rearranging terms, we have

$$\Big(\sum_{n=1}^{N}Q^n_{T-1}\Big)^2 \le \frac{8N^2D^2}{\mu}T\ln(Te) + \frac{4N^2D^4}{\mu^2}[\ln(Te)]^2. \tag{40}$$

Taking the square root on both sides of (40) and applying $\sqrt{a^2 + b^2} \le a + b, \forall a, b \ge 0$, we have an $\mathcal{O}(\sqrt{T\log T})$ original virtual queue bound

$$\sum_{n=1}^{N}Q^n_{T-1} \le \frac{2\sqrt{2}ND}{\sqrt{\mu}}\sqrt{T\ln(Te)} + \frac{2ND^2}{\mu}\ln(Te). \tag{41}$$

Combining (41) with $\text{VIO}(T) = \frac{1}{\beta}\sum_{n=1}^{N}Q^n_T$ in (6) and $Q^n_T \le Q^n_{T-1} + \beta G, \forall n$ from (5) under $\beta = 1$, we have (22). ■

Appendix I: Proof of Theorem 6

Proof: Substituting $f_t(x^\star) - f_t(x_t) \le RD, \forall t$ and $Q^n_{T-1} \ge 0, \forall n$ into (20), we have $\sum_{n=1}^{N}\sum_{t=1}^{T-1}\hat{Q}^n_t[g^n_t(x_{t+1})]_+ \le \frac{4ND^2}{\mu}T\ln(Te) + \frac{2ND^4}{\mu^2}[\ln(Te)]^2$. Combining the above inequality with (37) in the proof of Theorem 3, and noting that $-\sum_{t=1}^{T-1}\hat{\Delta}_t \le 0$ below (38), we have

$$\begin{aligned}\sum_{t=2}^{T}\sum_{n=1}^{N}[g^n_t(x_t)]_+ \le& \frac{G}{V\epsilon}\sum_{t=1}^{T-1}\sum_{n=1}^{N}\max_{x\in\mathcal{X}}|g^n_{t+1}(x) - g^n_t(x)| \\ &+ \Big[\frac{4ND^2T\ln(Te)}{V\mu} + \frac{2ND^4[\ln(Te)]^2}{V\mu^2} + \frac{2NG^2}{V}\Big]. \end{aligned}\tag{42}$$

Letting $V = \frac{GT\log T}{2}$ and $\epsilon = \frac{1}{T\log T}$ to bound the last term on the RHS of (42) by $\mathcal{O}(1)$, and noting that $[g^n_1(x_1)]_+ \le G$ from Assumption 3, we have (23). ■

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